

Networked information management

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Every day millions of people trawl the Internet for information using any one of a dozen or more different search tools. Whether they find what they are looking for may depend not only on their skill, but also on their luck. In the corporate arena, organisations are making increasing amounts of information available via intranets. This paper looks at the limitations of current networked information management technology, in particular, shortcomings in the areas of retrieving, organising and sharing of information, and an information management process which would overcome these problems is described. The representation of a user's information needs and interests in a user profile is seen to be central to the process and work in this area, including a novel, non-explicit approach to the representation of profiles, is covered. Four information access systems developed at BT Laboratories are discussed and the extent to which these are currently able to support the information management process is considered.

1. Introduction

Every day millions of people trawl the Internet for information using any one of a dozen or more different search tools — but whether they find what they are looking for sometimes depends not only on their skill, but also on their luck. Studies indicate that many existing search engines do not meet the needs of users [1].

Recently, there has been an increasing interest both in the use of Internet technology within organisations and in the emergence of intranets [2]. More and more corporate information is being stored in intranet networks and this has led to the realisation that these intranets are valuable repositories of corporate knowledge. Organisations are increasingly realising the importance of using knowledge and information residing in their networks for competitive advantage:

‘Intranets are providing many companies with cheaper, easier to install, and more flexible private networks for distributing text, graphics, and video to geographically distant workers. Sales of Intranet software will increase from \$142M in 1995 to \$1200M in 1997’.

Wall Street Journal, 1995.

Key to the usefulness and viability of both the public Internet and corporate intranets, however, will be the ability to manage information, and provide users with the information they want, when they want it.

This paper firstly looks at the limitations of current information management technology, and, in particular, at shortcomings in the areas of retrieving, organising and

sharing of information. An information management process is described which would overcome these problems. The representation of a user's information needs and interests in a user profile is seen to be central to the process and work in this area is covered in section 4, including a novel, non-explicit approach to the representation of profiles. Four information access systems developed at BT Laboratories (BTL) are then discussed, and the extent to which the information management process can currently be supported with computer-based tools is considered. Finally, further work to produce an information agent capable of supporting the process in its entirety on the user's behalf is briefly proposed.

2. Current technology — some limitations

There are a number of problems encountered with current information management technology.

Most problems centre around difficulties in finding relevant information. Simple keyword queries often yield too many results of variable quality. It is usual to receive several hundred hits (documents matching the user's query) even on relatively narrow queries and to receive thousands of hits for a query is not uncommon. Research also shows that users do not find it easy to frame the query needed to return the information they require [1].

Text retrieval systems can be characterised by two measures, recall (*R*) and precision (*P*). Recall measures the retrieval effectiveness of the system; the number of documents which are actually retrieved compared with the total number of relevant documents in the database:

$$R = \frac{\text{number relevant retrieved}}{\text{total number relevant}}$$

It has long been realised that the recall of text retrieval systems can be quite low. One study [3] showed that less than 20% of the relevant documents were retrieved from a database containing roughly 350,000 pages of text, using IBM's STAIRS full-text retrieval system. Interestingly, the subjects involved in the trial believed that they were retrieving a much higher fraction of the relevant documents. The study also claims that evaluations done on small databases do not scale to large databases, since the amount of search effort required to obtain the same recall level increases as the size of database increases, often at a faster rate than the increase in database size. This has obvious implications for the very large information sources which are now becoming available on the Internet.

Precision (P) measures how relevant the retrieved documents are to the query:

$$P = \frac{\text{number retrieved relevant}}{\text{total number relevant}}$$

A reasonable precision is important if the user is not to be deluged with irrelevant documents, which can be highly time consuming and frustrating for the user.

Unfortunately, recall and precision are mutually dependent. Figure 1 is typical of the performance of a traditional text-retrieval system. It shows clearly the trade-off between recall and precision. If a search aims for high precision then only a small fraction of relevant documents are retrieved. On the other hand, a search with high recall also finds a large number of irrelevant documents which must be discarded. In the limit of 100% recall, the precision has dropped to zero, which means that all of the relevant documents can only be found by retrieving the entire database.

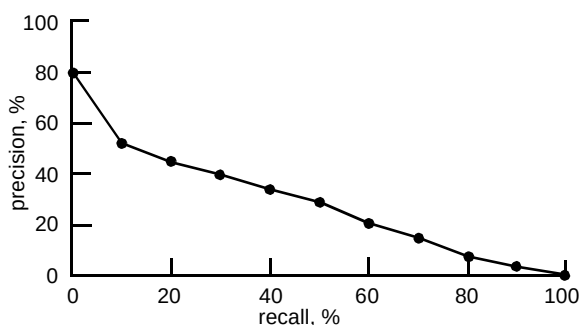


Fig 1 Trade-off between recall and precision.

The performance of current systems is disappointing even when used on single databases. When searching multiple networked information sources, a number of additional effects come into play, including repeated documents, multiple slight variations of essentially the same information and the problem of information granularity —

current systems have no way of differentiating between detailed documents and summaries or overviews.

In addition to these problems, organising information once it has been found may not be straightforward. Most current WWW browsers (Netscape Navigator, Internet Explorer, etc) incorporate a bookmark facility to provide some means of storing pages of interest to the user. Typically, this is done by allowing the user to create a hierarchical menu of names associated with particular URLs. While this menu facility is useful, it quickly becomes unwieldy as the number of bookmarks grows. Essentially, the representation provided is not rich enough to allow capture of all the relevant data about the information stored — the user can only provide a string naming the page. As well as the fact that useful meta-information such as the date of access of the page is lost, a single phrase (the name) may not be enough to accurately index a page in all contexts. Consider, as a simple example, information about the use of knowledge-based systems (KBS) in information retrieval of pharmacological data — in different contexts, it may be KBS, information retrieval or pharmacology which are of interest. Unless a name is carefully chosen to mention all three aspects, the information will be missed in one or more of its useful contexts. This problem is analogous to the problem of finding files containing desired information in a computer file system [4], although in most such systems one can, for example, sort files by creation date.

Lastly, the sharing of information is not supported in most current systems. In the corporate context, when one user finds information which is important or relevant there will often be other users who also need to know — for example, other members of a project team. Ideally, this kind of information sharing should be automated using software which knows the interests and role of each user.

Given the increasing size and availability of information sources and the limitations of current systems discussed above, novel techniques will be required for information retrieval in the future. These techniques will need to be:

- appropriate for both textual and non-textual information,
- designed with networked retrieval in mind,
- able to scale up to deal with increasingly large information stores,
- capable of extracting information from documents at appropriate levels of detail,
- able to model users' interests and able to find and share relevant information.

3. The information location and management process

Information sources are increasing in size, complexity and number and, as shown in the previous section, current information retrieval techniques are insufficient for very large, multi-modal, networked information sources. The Internet is an early example of the information networks of the future and is increasingly being commercially exploited.

Because of the large amount of information available, user profiling will be crucial to the information products and services of the future. 'Mass customisation' — the possibility of customising the service to each customer based on that customer's priorities and interests — could be a key market differentiator.

A number of 'point solutions' have been developed, addressing some of the technology challenges mentioned above. These include the Jasper WWW information agent, the ProRate user profiling software, ProSearch, an agent for automatically searching WWW on the user's behalf, and the nDir automatic categorisation tool. These systems will be described in more detail in later sections.

A more complete solution would involve the construction of an integrated set of information tools to support the information location and management process. This process can be divided into four main parts, not all of which will necessarily be applicable in every situation (see Fig 2).

The remainder of this section looks at each stage of the process in turn.

3.1 Searching

A search query may comprise a set of keywords provided by a user or it may be automatically generated. Examples of automatically generated queries are those generated from a user's profile (perhaps to periodically search for items of interest to that user) or from a document (typically to find similar documents).

Once a query has been created, it can then be enhanced. One way to do this is to use a thesaurus to add to the words in the query synonyms or other related words (e.g. meronyms). An alternative approach is to use keyword clustering, a technique which statistically analyses a set of documents and discovers which keywords are most likely to co-occur. Given a keyword in a query, related keywords can then be added to the query. The assumption here is that co-occurrence is an indicator of the degree to which keywords are related. This approach has the advantage that if the corpus (set of documents) chosen is from the same general area as the queries being formulated, problems such as multiple meanings associated with thesaurus-based approaches can be mitigated to some extent.

In many cases, particularly with large and diverse information sources such as the Internet, too many hits are returned. Furthermore, these hits are of variable quality. Factors affecting the results returned from a search include:

- users, and their ability to frame a query which will return documents matching their information needs,
- the information source (as mentioned above, the Internet contains tens of millions of documents of highly variable quality, thus making it a difficult corpus to search),
- the quality of the algorithms used in the search software ('search engine').

Very often, therefore, the results a search engine returns need to be refined and filtered before presenting them to the user.

3.2 Refinement

Refinement of a set of documents (typically the result of a search) can be carried out in a number of ways. Obviously, in order to refine the search some new information is needed, which can be exploited to give a clearer view of the documents likely to be relevant. This additional information can be in several forms, for example:

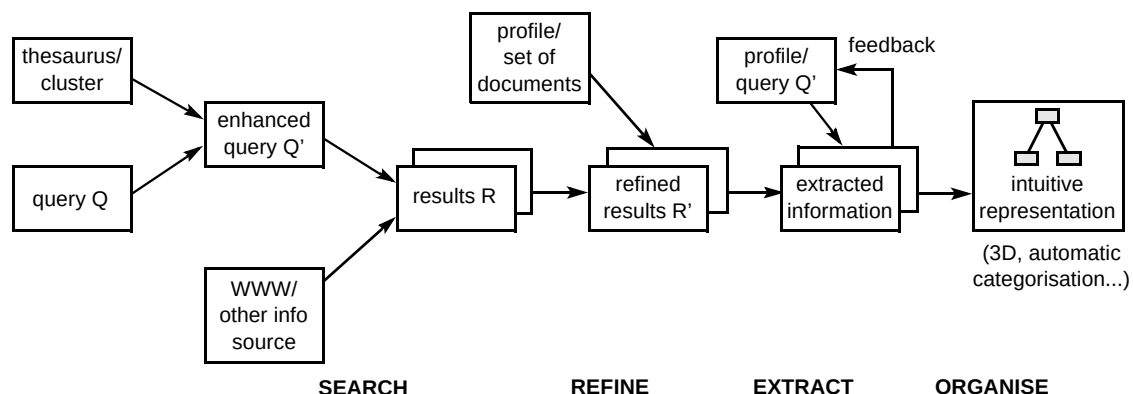


Fig 2 The information management process.

- a user profile,
- a document or set of documents in which the user is known to be interested,
- an implicit profile of the user's interests from which their actual profile can be extracted (e.g. a WWW bookmarks file).

A number of techniques can then be used to refine the initial search results. Two particular techniques are:

- clustering where algorithms take a set of documents and organise them into related groups — here the set of documents to be clustered would be the search results plus some representation of the user's profile or a set of documents in which the user is known to be interested, documents emerging in the same cluster(s) as the user's profile or representative documents being likely to be most relevant,
- secondary searching — in this approach, each document retrieved is matched against the user's profile (or set of representative documents), with close matches most likely to be relevant.

Use of these refinement and filtering techniques will yield a more accurate, higher quality list of relevant documents.

3.3 *Extraction*

It is important to remember that users are typically looking for information not 'documents'. The information a user requires may be contained in a larger document or indeed in a number of segments of more than one document. The information extraction task aims to create abridgements of one document or a digest from a set of documents to pull out from the document(s) the information for which the user is looking.

Abridgements and digests can be general in nature, i.e. they attempt to provide an abridgement of the information contained in a document or set of documents, or they can be more specialised in a variety of ways:

- the original user query can be revisited and the parts of the document(s) most relevant to the query can be extracted,
- a digest can be created based on a user's profile,
- information can be extracted which is most related to a given document or set of documents.

The use of information extraction has a number of benefits. They allow a user to access the information in which they are most interested without having to read a whole document or set of documents; abridgements or digests of remote information can be stored locally — this abridgement of information can then be accessed very quickly before making a decision to retrieve lengthier remote documents; digests can remove the need to read more than one document by gathering together the key points from a number of documents into a single local document.

When information is presented to users, they can indicate its relevance. The system can then use this information to update any user profile to better reflect the user's interests as reflected by the feedback they have provided [5].

3.4 *Organisation*

Having found relevant documents and extracted the information requested, the requirement then is to make the information easily accessible in future, both to the user and possibly to others.

The clustering techniques mentioned above can be used to automatically categorise information found into related topics for easier future access. The set of documents can also be represented in 3-D which can aid browsing and searching for documents [6].

When useful information is found, there will often be a desire to share it with colleagues or others; this can be automated, e.g. the Jasper information agent [7,8] will automatically e-mail users when pages of interest to them are found.

3.5 *Push and pull models of information access*

Until recently, information has been obtained from the World Wide Web via a request and reply mechanism, or 'pull' model — the client initiates a request for information and the server replies. In some circumstances, however, the user wants to be notified when the data has changed. This requires a 'push' information model. The user subscribes to information by preselected topic areas and as information becomes available on the information provider's server that matches the user's interests it is pushed out to the client. This model has obvious advantages in areas where information needs are well known and easily defined and where information may be changing rapidly. Examples of systems which operate on the push model are BackWeb [9]

and Castanet [10]. Cochrane [2] looks at push information models in more detail.

In fact, 'push' is currently something of a misnomer in most cases. Many push systems actually hide the 'pull' from the user by scheduling periodic requests from the client to the server. The pull still takes place but need not be explicitly requested by the user. The push model could thus perhaps be more accurately renamed the 'automatic periodic pull model'.

The main reason the pull remains is that any information downloaded from the Internet has to take a specific route across the network — the client machine is a node on the network and requires a one-to-one link to the server machine. If the link is not established, perhaps because the client is not connected to the network, then the server is unable to send anything to the client. Hence a client-side request has to be made to the server, albeit automatically. Research is ongoing, however, into reliable multicasting techniques which would deliver true push technology [2].

Whether information is pushed or pulled, however, the requirement to identify and refine information relative to a particular user's interests is still a key issue. Typically, current push information services use manual (human) information selection and management techniques. The amount of information that can be processed in this way is clearly limited and automation of the techniques described above would be an obvious benefit.

4. Personalisation — user profiling and information filtering

As shown above, one key to effective information management is user profiling — the ability to represent and reason about the interests or preferences of a user. Use of user profiling techniques will allow software to automatically filter and prioritise information from, for example, collections of Internet/intranet multimedia documents or in the future from several hundred channels of digital TV.

This section will discuss techniques for representing users' interests and filtering the vast amount of information that is now available from the Internet (and other electronic sources) in a variety of media. The current maturity of each technique is considered and different techniques for different application types are recommended.

There are three different approaches to user profiling:

- statistical keyword-based approaches,
- artificial intelligence (AI), and neural network (NN) techniques,

- social filtering (a non-explicit approach, using the preferences of other users rather than an explicit representation of a profile).

4.1 Statistical keyword approaches

Keyword matching

Keyword (or term) matching is perhaps the best understood approach to determining whether some information is relevant to a user's interests. A common method for implementing a keyword matching scheme is to use a vector space model (VSM). Such a model constructs a vector of dimension m for each document, where m is the total number of terms (keywords) used to identify the content of the documents in the system. There are three main approaches to producing a document's vector:

- Boolean — if the term appears in the document then it is given a weighting of 1, otherwise it is weighted as zero,
- term frequency where the weighting given to each term is based upon the frequency of its occurrence in the document — if the term is not present, then it is set to zero, the resulting vector often being normalised to prevent longer documents from having a better chance of retrieval,
- TF-IDF — the weighting given to a term under this scheme depends on the term frequency (TF) (which is the frequency of the occurrence of the term within the current document), and the inverse document frequency (IDF), the IDF being a measure of how often the term appears in all of the documents in the retrieval system and often calculated as :

$$\text{IDF} = \log(N / n)$$

where N = total number of documents in retrieval system ,
 n = number of documents containing the term

Hence common terms have a low IDF and terms that appear in only a few documents will have a high IDF. The weighting for any particular term in the document is then given by:

$$w = \text{TF} \times \text{IDF}$$

As with the term frequency approach, normalisation of the document vector is common, as is matching against phrases rather than individual terms.

A user's profile can be mapped into the vector space and the similarity between a document and the profile is then calculated as the cosine product of the two associated vectors. If the result is greater than the pre-defined

relevance threshold for that user profile then the document is presented to the user.

The advantages of this approach are its relative simplicity and its adoption of well-understood and effective techniques from the field of information retrieval [11]. Disadvantages include the fact that the technique can only be used for text documents. Additionally, as there are many ways to express a given concept (synonymy), the literal terms in a user's profile may not match those in a relevant document. Conversely, as most words have multiple meanings (polysemy), terms in a profile may literally match those in irrelevant documents. These problems can, however, be mitigated somewhat by the use of profile enhancement (see below). Generally, this scheme (incorrectly) assumes keywords are orthogonal in vector space, i.e. all terms are unrelated. The idea of synonymy shows that several keywords could describe the same concept. Such keywords are clearly related and therefore cannot be classed as orthogonal. On a less obvious scale, related concepts are also not orthogonal (see the next section for further discussion on this point).

Profile enhancement

A supplementary method for improving the performance of keyword-based filtering adapts the information retrieval technique of query expansion. The filtering system expands the profile by adding terms related to the keywords supplied by the user. These new terms can either be derived with lexical aids, such as thesauri and controlled vocabulary lists, or statistically selected based upon the general co-occurrence of the terms in other documents (see section 5.2).

The procedure used in Voorhees [12] relied upon sets of strict synonyms (known as synsets), which formed part of a lexical system called WordNet [13]. The synonyms are arranged into 'is-a' and 'part-of' relationships. The expansion of any given query could then proceed in several ways by adding (for example):

- only the immediate synonyms from appropriate synsets,
- all descendants in the is-a hierarchy,
- all words in synsets which are one is-a or part-of link away from the original synset.

It was found that the expansion technique did not improve the effectiveness of queries that were relatively complete when entered by the user, for any filtering method. However, shorter or incomplete queries were significantly improved by the expansion and these still occur relatively often.

4.2 Artificial intelligence and neural networks

Learning information agents

Learning information agents infer the preferences of their users by using feedback to update the user profiles. Some of the applications of this method described in the literature [14] involve keyword searching the WWW for pages that might be of interest, and then applying a filter to the results, presenting only the top p pages to the user. This can easily be adapted to include a relevance threshold so that whenever a new page is considered it can be rated and presented immediately, if relevant. Alternatively, the predicted score for the document can be fed back to the user.

The normalised TF-IDF weighting system is used (see section 4.1) to create a vector, V_i , for the new document. User profiles are also maintained, based on the same terms. This takes the form of a vector, M (of the same type as V_i), where the weight of each term corresponds to the user's preferences. A document is then scored (as with the vector space model), by simply taking the dot product $V_i \cdot M$. The relevance ratings that a user can give to a document should cover a range of values, for example -5 to $+5$.

The information supplied to the learning part of the system consists of a selection of documents and the associated evaluations given by the user. The learning algorithm uses this information to update the weights in the user's profile. A decay function may also be applied so that terms that have not appeared in recent documents deemed relevant by the user will start to lose their weighting. This may be useful for matching changes in the user's interests over time.

A good way to initialise the user profile is to ask a new user to decide whether each of a selection of existing documents is relevant or not. It would also seem to be more effective to have a relatively low relevance threshold initially, and increase it as the system learns more about that user. This ensures that the user gets to view pages that may normally be classified as irrelevant and has the chance to indicate their interest if appropriate.

The weights can be updated by a simple addition process. Assume the user gives an evaluation e_i to page i . Then the new user vector (i.e. profile), M^+ , becomes:

$$M^+ = M + \sum_{i=1}^p (e_i \times V_i)$$

where p is the number of pages evaluated.

Neural networks

Neural network agents also learn the interests of the user with continued use of the system. A user profile consists of a neural network [15] which represents the concepts which are important to that user. The network is based on terms which occur in those documents that are accepted and those that are rejected when presented to the user as a possible result of a query they have made. The network also models the associations between words, so that related words are related in the network. For example, if the user has read many documents related to 'telecommunications' and a new document uses the terms 'telephone communications' then there is a high probability that the association will be made by the network [16]. This occurs because the terms 'telecommunications', 'telephone' and 'communications' will have relatively higher interconnection weights, hence indicating the association. This is due to these terms appearing together in several of the documents processed by the network.

If a store of the documents is maintained, then enhanced keyword searching can be performed. This is achieved by using keywords to search the store and the user profile to rate the documents found.

New significant words (e.g. derived from TF-IDF weights) that appear often in documents accepted and read by the user, can be added to the network for a more comprehensive profile, or suggested to the user for approval of the addition. This technique can also be used to initialise the network, or the user may be prompted to manually enter a selection of keywords of interest.

This approach copes automatically with changes in users' interests and maintains a changing and more accurate user profile. The user does not have to decide on a score — they simply accept or reject each document that is presented to them. As with keyword weighting, these techniques can only be used for text documents.

4.3 Social filtering

Filtering by recommendation

The user profiling approaches examined above have all been based on the keywords or terms found in documents and user profiles. These approaches can be described as content-based filtering. However, there are two major disadvantages associated with all content-based filtering techniques. Firstly, the items that are being filtered must be text documents or some other form that can be parsed by a computer. Hence items such as sounds, video clips, images, and so on, cannot be easily filtered with these systems. Secondly, content-based filtering has no inherent method for presenting the user with items that they did not know

that they liked, since the filtering only selects items that fit the defined user profile.

Social filtering overcomes these problems. It enables a user to filter the information that they receive based upon the ratings given by other users in the system. This often relies upon collaborative agents where the user agents work together to produce recommendations. Generally, social filtering automates the process of 'word-of-mouth' recommendations but uses the ratings of many users to make these recommendations more accurate to a particular user's tastes.

Social filtering can also lead to the recommendation of items in a category that the user has not previously declared as interesting or relevant — hence eliminating the second disadvantage above. This can arise when other users, with similar interests or tastes (in other categories) give a high rating to an item in this new category.

A user's profile is simply the set of ratings that they have given to a selection of items from the system database. As can be expected, the greater the number of ratings in a user's profile, the more accurate the recommendations will be for that user.

The filtering is based upon feedback from other users who have already rated an item that is unrated by the current user, either implicitly or explicitly. Implicit rating may be as simple as sending a reply to a news article. Explicit feedback involves the user actively giving an item a score based on their opinion of it.

The use of the feedback then depends on its type. For implicit Boolean feedback, it could be: 'Show me all news articles to which Fred Smith made a reply.' Alternatively, when the feedback involves explicit ratings, an ordered list of recommendations with predicted scores may be generated.

Several applications of social filtering can be found in the literature [17].

Algorithms

There are several algorithms that have been used in social filtering systems, four of which are discussed briefly here.

The first is to simply set the predicted rating of an item to the average score for that item. This technique cannot be expected to produce highly accurate results but is a good algorithm against which to compare others.

A more mathematical approach is to find a user with a similar profile of ratings. This is actually calculated as the (lowest) dissimilarity which is represented by the mean

squared difference between two profiles. The ratings from the most similar user can then be used to generate the recommendations. There is no reason why the recommendations cannot be based upon the ratings of several similar users.

A third algorithm that can be used is the Pearson r algorithm. The Pearson r correlation coefficient can be taken as a measure of similarity between two user profiles. Each user profile consists of the ratings that the user has given to a selection of items.

The final algorithm to be considered here is that used in BT's ProRate agent and MORSE film recommendation system [18]. To estimate the rating that user i would give film k , the following algorithm is used:

Calculate the correlation between k and every other film, (based on ratings).

For every other user j do :

Determine the ratings given by j to the N films most closely correlated with k .

Determine the correlation between i and j , (based on ratings).

End For

Extrapolate estimated rating for film k by user i .

(N is varied to achieve the best results.)

This algorithm generates a hypothetical user with exactly the same tastes as the current user and can hence give a rating for all films in the database, but particularly those with sufficient ratings to be statistically accurate. Full details of this algorithm can be found in Fisk [18].

These four social filtering algorithms have been implemented in order to compare the results. As discussed

elsewhere [19], the ProRate personal recommendation agent gives the best results.

A major advantage of the social information filtering approach is that it is not limited to text documents but can be used on any user rateable item. It becomes more accurate as the user profile is extended and may generate recommendations for items that a user did not know they wanted. To be most effective, a relatively large number of ratings from several users is required [20].

5. Case studies

It was mentioned in section 3 that BT has developed a number of systems to address different phases of the information discovery and management process described there — a number of these systems are now discussed here.

5.1 Jasper

Jasper [7, 8] is a system of World Wide Web (WWW) information agents. Figure 3 summarises the main functionality provided by the Jasper information agent.

Storing information in Jasper

Jasper agents are used to store, retrieve, summarise and inform other agents about information found on WWW. In certain circumstances, Jasper agents will also identify an opportunity for performance improvement and will seek user feedback in order to improve. Each Jasper user has a personal agent which holds a user profile based on a set of key phrases.

Given the vast amount of information available on WWW, it is preferable to avoid the copying of information from its original location to a local server. Indeed, that approach, it could be argued, is contrary to the whole ethos

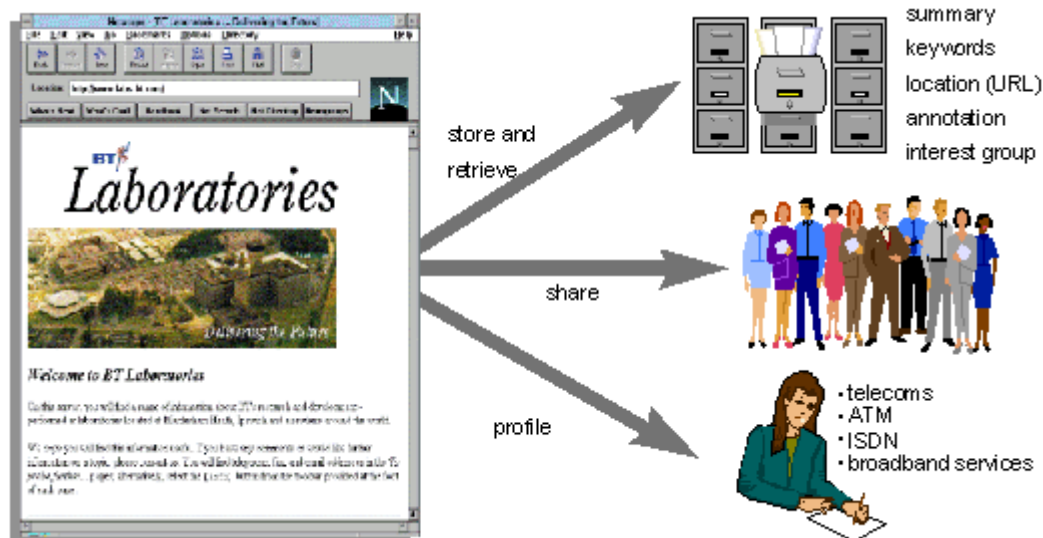


Fig 3 The Jasper information agent.

of the Web. Rather than copying information, therefore, Jasper agents store only relevant meta-information. This meta-information is then used to index on the actual information when a retrieval request is made.

When users find a WWW page of sufficient interest to be stored by Jasper, a 'store' request is sent to Jasper via a menu option on their WWW client. Jasper then invites the user to supply an annotation to be stored with the form. Typically, this might be the reason the page was stored and can be very useful for other users in deciding which pages retrieved from the Jasper store to visit. The user can also specify at this point one of a predefined set of interest groups to which to post the page being stored. The WWW page is added to the Jasper store. Essentially, the Jasper store is a simple term-document matrix M , wherein:

$$M(i, j) = \begin{cases} 0 & \text{if term } j \text{ does not occur in document } i \\ n & \text{if term } j \text{ appears in document } i \text{ } n \text{ times} \end{cases}$$

At storage time, Jasper agents perform four tasks:

- an abridgement of the textual information on the WWW page is created, to be held locally, thus enabling a user to quickly assess the content of a page from a local store before deciding whether to retrieve remote information,
- the content of the page is analysed and matched against every Jasper user's profile — if the profile and document match strongly enough, Jasper e-mails the user, informing him or her of the page which has been stored,
- the page is also matched against the storer's own profile — if the profile does not match the content of the page being stored, Jasper suggests to the user new words or phrases they may wish to add to their profile, and in this way Jasper agents have the capability to adaptively learn the user's interests by observing their behaviour,
- for each document, an entry in the Jasper store is made, holding keywords, an abridgement of the document, document title, user annotation, universal resource locator (URL), storer name, and date and time of storage.

In this way, a shared and enhanced information resource is built up in the Jasper store. Given that users must make a conscious decision to store a page, the quality of the information in the Jasper store is high — it is effectively pre-filtered by Jasper users.

Getting information from Jasper

E-mail notification — as described above, when a page is stored by a Jasper agent, the agent checks the profiles of other agents' users in its 'local community' (the set of users who contribute to that particular Jasper store). If the page matches a user's profile with a score above a certain threshold, an e-mail message is automatically generated by the agent and sent to the user concerned, informing the user of the discovery of the page.

What's new — a user can ask his Jasper agent 'What's new?' The agent then interrogates the Jasper store and retrieves the most recently stored pages. It determines which of these pages best match the user's profile. A WWW page is then presented to the user showing a list of links to the recently stored pages, along with annotations where provided, date of storage, the storer, a link to a locally held abridgement of the page and an indication of how well the page matches the user's profile.

Interest groups — as mentioned above, when a user stores a page in Jasper, he has an opportunity to specify one of a predefined set of interest groups to which to post the page. Interest groups gather together pages of related information. Jasper users can visit interest group pages which are dynamically constructed from the pages which have been posted to them and consist of a list of links to the pages and their abridgements, along with any annotation provided by the original storer of the page. Interest groups are similar to the 'list of links' pages found in many WWW locations, with the important extensions that multiple users can contribute to the list (automatically via the storage process) and that abridgements of the information and annotations by the original storer of the link are also available.

Keyword retrieval — the user supplies a query in the form of a set of keywords. The Jasper agent then retrieves the most closely matching pages held in the Jasper store, using a Boolean keyword matching and scoring algorithm. The agent then dynamically constructs an HTML page with a ranked list of links to the pages retrieved and their abridgements. Any annotation made by the original user is also shown, along with the scores of each retrieved page. This page is then presented on the user's WWW client.

Show me similar — this facility allows a Jasper user to request that his agent retrieves relevant pages from the Jasper store. The WWW page currently being viewed by the user is analysed and the Jasper store is searched for related documents. Given the pre-filtered nature of the Jasper store, it provides a rich resource for this type of facility.

5.2 ProSearch — a proactive search agent

Clustering keywords

Keywords (terms) occurring in a particular document collection can be clustered into related groups. Consequently, a similarity matrix for the keywords in the Jasper store (or some other document collection), for example, can be constructed which gives a measure of the ‘similarity’ of each pair of keywords in the store. For each pair of keywords, the Dice coefficient is calculated [21]. For two keywords K_i and K_j , the Dice coefficient is given by:

$$S_{K_i K_j} = \frac{2C}{A+B}$$

where

$S_{K_i K_j}$ = similarity between K_i and K_j

C = number of documents in which K_i and K_j co-occur

A = number of documents in which K_i occurs

B = number of documents in which K_j occurs

A similarity S of 0 means that two terms never co-occur, while if S was 1, the terms would occur in the same and only the same documents.

Note that this formula assumes Boolean document weighting — that is, the weight of a given document D is 1 if the keyword occurs in D , else 0. More sophisticated schemes based on keyword frequency are available.

Proactive searching

The keywords comprising users’ Jasper profiles are being used by an experimental system called ProSearch to

search proactively for new WWW pages relevant to their interests, which can then present a list of new pages in which the users may be interested, without them having to explicitly carry out a search (see Fig 4). These proactive searches can be carried out at some given interval, perhaps weekly or monthly. Clustering is useful here because a profile may reflect more than one interest. Consider, for example, the following user profile:

Internet, WWW, html, http, football, Manchester, United, linguistics, parsing, pragmatics.

Clearly, three separate interests are represented in the above profile and searching on each separately is likely to yield far superior results than merely entering the whole profile as a query for the given user. Clustering keywords from the document collection will automate the process of query generation for proactive searching by a user’s ProSearch agent.

Talking, for example, the following set of keywords as a typical profile:

{internet, information, retrieval, SMART, clustering, agent, intelligent, CORBA, IDL, DCE}

These keywords can be clustered as described above, using the Jasper store as the document collection, from which can be obtained the similarity matrix shown in Table 1.

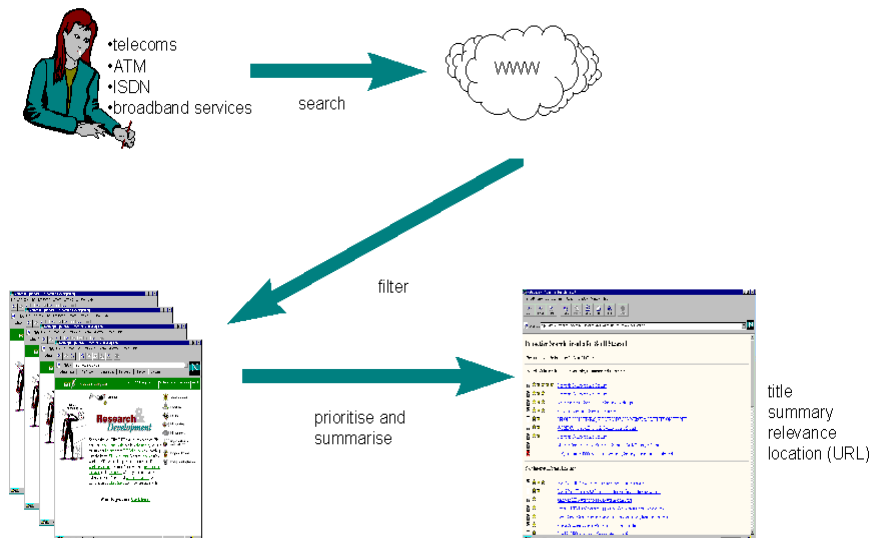


Fig 4 The ProSearch agent.

Table 1 A term similarity matrix.

	corba	dce	idl	internet	information	retrieval	intelligent	agent	SMART	clustering
corba	1									
dce	0.62	1								
idl	0.92	0.50	1							
internet	0.03	0.03	0.03	1						
information	0.04	0.04	0.03	0.51	1					
retrieval	0.04	0.08	0.00	0.21	0.27	1				
intelligent	0.00	0.03	0.00	0.21	0.30	0.30	1			
agent	0.07	0.11	0.04	0.16	0.29	0.38	0.53	1		
SMART	0.00	0.00	0.00	0.07	0.07	0.19	0.10	0.14	1	
clustering	0.00	0.00	0.00	0.01	0.03	0.17	0.04	0.12	0.43	1

If complete-link clustering [22] is used, whereby the similarity between the least similar pair of items from two clusters is taken as the similarity between the clusters, we obtain the cluster dendrogram shown in Fig 5. Note that the vertical axis in Fig 5 represents difference rather than similarity. If we set a similarity threshold of 0.3, below which any clusters are disregarded, the following clusters are obtained:

{dce, corba, idl}
 {SMART clustering}
 {internet information}
 {intelligent agent}
 {retrieval}

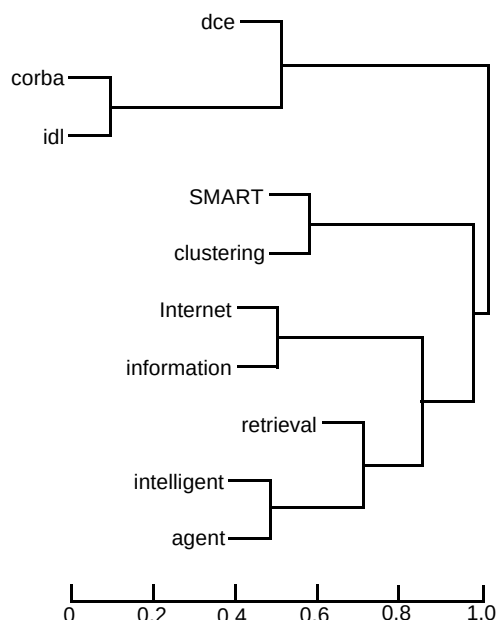


Fig 5 Clustering dendrogram.

Inspection of the results of clustering this and other profiles indicates that clustering is able to subdivide a Jasper profile into sets of keywords which may prove valid for use as queries. More work remains to be done, however,

to determine the best clustering technique (single-link, complete-link or group average) and the most appropriate clustering threshold for a wider range of profiles. It may prove necessary to attempt to dynamically tune the threshold (and indeed perhaps the clustering technique) based on the similarity matrix for a given profile. Consideration is also being given to a semi-automatic technique, whereby the set of queries created by ProSearch from the user's profile is presented to the user for confirmation or adjustment before any queries are actually submitted.

Because queries based on profile clusters are made off-line, rather than interactively by the user, there is an opportunity to analyse the results before presenting them to the user. For example, when the search results are obtained, they can be abridged and matched against the user's profile in the usual way to give a prioritised list of new URLs along with locally held abridgements. In addition, a log can be held of the pages which Jasper has found for each user. By checking against this log, only new pages (those the user has not been shown by ProSearch previously) will be presented. The pages will be presented to the user on a dynamically constructed WWW page with the facility to easily store pages of particular interest in the Jasper store. Those pages accepted for storage in Jasper by this mechanism (and indeed those rejected) may additionally afford new opportunities for user profile adaptation [5].

Profile enhancement

ProSearch is also being used to enhance the profiles of users of the Jasper system (see section 5.1) and other information filtering systems within BT. As discussed in 4.1, a user's profile can be enhanced by adding those keywords most similar to the keywords explicitly represented in the user's profile, in a way reminiscent of query reformulation techniques, such as the use of spreading activation networks [23, 24] and other techniques based on the use of semantic nets [25, 26].

In Chakravarthy and Haase [25], for example, the use of semantic knowledge from WordNet [13] in an experimental

program called NetSerf for finding information archives on the Internet is reported. Based on a set of 75 queries, NetSerf was reported to have performed significantly better than the Smart [27] information retrieval system. Interestingly, some simple experiments with WordNet have been performed at BTL for profile enhancement, but it was found that WordNet did not perform well with respect to the very technical vocabulary which constitutes the terms in the Jasper store. It was this technical bias that led to looking to the store itself for a network of related terms — in effect, the clustered terms constitute a simple spreading activation network. In Ruge [23], some results from cognitive psychology research are analysed to explain why it proves difficult for people to think of related terms when trying to frame a query to an information retrieval system. This difficulty is ameliorated to a degree by the automatic (hidden) enhancement of a user's profile by the system.

Figure 6 shows an example network of keywords which have been built from the keyword similarity matrix extracted from the Jasper store. The algorithm is straightforward — given an initial starting keyword, find the four words most similar to it from the similarity matrix; link these four to the original word and repeat the process for each of the four new words. This can be repeated a number of times (in Fig 6, three times). Double lines between two words indicate that both words occur in the other's four most similar keywords. One could of course attach the particular similarity coefficients to each link for finer-grained information concerning the degree of similarity between words.

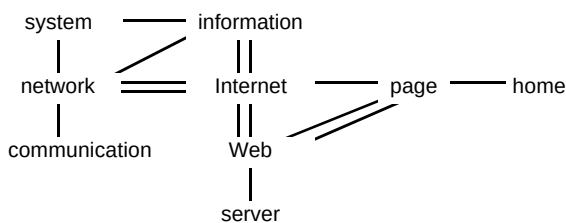


Fig 6 A keyword network from Jasper.

Thus, for example, if the words virtual, reality and Internet are part of a user's profile but VRML is not, an enhanced profile might add VRML to the original profile (assuming VRML is clustered close to virtual, reality and Internet). In this way, documents containing VRML but not virtual, reality and Internet may be retrieved whereas they would not have been with the unenhanced profile.

5.3 nDir

Automatic categorisation of on-line information

Current WWW indexes generally fall into two categories — automatically generated indexes of document

keywords, such as Alta Vista [28], and manually generated hierarchies of categories, such as Yahoo [29].

Keyword indexes are easily generated and updated but are difficult for naive users to use. The only way to access the index is to perform a keyword search and the user must understand how to choose keywords and construct queries that will return good-quality results.

Categorised material is easier to use since locating information is more intuitive than when keyword searching. A site such as Yahoo allows the user to navigate through a series of menus and submenus until they find the information they want. The placing of a page within the directory is generally done manually — someone has to decide in which category a page should belong. If the category structure has to be changed then pages would need to be categorised again, making updating the directory a laborious process. Having several category structures for a single set of pages would make this problem even worse.

nDir (Network Directory) brings together the best of automated indexes and easy-to-use directories. The nDir system can categorise sets of pages automatically, based upon predefined keyword profiles.

Technical details

There are three stages to building an nDir system — defining the profiles, scoring the pages and building the category structure.

A profile is simply a set of keywords which are associated with documents on a particular subject. With each keyword is a weighting from 1 to 5 depending on how strongly that word places a page within the subject area. Profiles are very important to the success of the nDir system — a profile containing well-chosen words will enable relevant pages to be scored very highly while irrelevant ones are filtered out.

Each WWW page has to be processed into a canonical form for scoring. This involves removing common words using a stop list and conflating words to their common stems. The stopping and stemming processes are well documented in the information retrieval literature [11] and involve, for example, the removal of plurals (-s, -es) and present participle (-ing) endings. The result is a list of words, each with a document frequency and a location marker to say if the word was found within HTML titles, headings or elsewhere.

Each page is then scored against the profiles. Initial experiments involved using TF-IDF weighting techniques (see 4.1), but this did not exploit the metainformation which is available about words in marked-up documents such as HTML pages — words found within HTML titles and

headings will, in general, have more relevance than words found in the rest of a document. Given this observation, a simpler formula than TF-IDF, which nonetheless took these word positions into account, was developed.

A category structure has to be created to define the 'category/sub-category' and 'see also' relationships. For example we have a category 'physics' as a sub-category of 'science' and a 'see also' relationship exists between 'physics' and 'astronomy'. A category has one profile or a combination of profiles associated with it. In the current versions of nDir there is generally one profile per category, i.e. there is a single 'physics' profile that corresponds to the 'physics' category.

The user can browse through the category structure and view the pages contained within each category. The pages are listed by score, with the most relevant pages for a category being viewed first. The user can also search for categories and individual pages by keyword.

Current status

A version of nDir currently exists for use in CampusWorld, an on-line information service for UK schools and colleges described elsewhere [30]. This version is intended to allow the users of CampusWorld to browse through and search the entire contents of the site easily. Results from categorising the CampusWorld content have been good, with users finding the facility useful in finding relevant information.

5.4 Remembrance Agent and RADAR

The Remembrance Agent (RA) [31] is a program which augments human memory, suggesting information which might be relevant to the user's current task. Unlike many other information-retrieval systems, the RA runs without user intervention. An unobtrusive interface allows a user to pursue or ignore the RA's suggestions as desired. The original proposal for and implementation of such a system are described in Jones [4].

Almost all current networked information-retrieval systems have concentrated on query-based information retrieval on demand. With increasingly powerful computers, however, much of a computer's processing resource can be spent waiting for user activity, the next arrival of network traffic, and so on. Thus there is an opportunity for the computer to exploit spare processing resource more constructively by performing continuous searches for information that might be of use in its user's current context. The Remembrance Agent performs this continuous, associative form of recall by continuously displaying relevant information.

RADAR is a reimplementation of some of the ideas of the RA. RADAR exploits networked information resources, including intranet documents, newsgroup articles and e-mail messages, by observing users as they create or read a document. RADAR extracts a query from keywords in proximity to the current cursor location in the document and searches its information sources using this query. As the document is created, read or edited, links to relevant information are displayed in a WWW browser window.

A version of RADAR is currently being trialled in a large software project which makes use of over 500 specification documents.

6. Conclusions

It is instructive to compare the information management systems described in the previous section with the features required to support the information management process, as described in section 3 (see Table 2).

Each of the systems examined contain some of the attributes identified as being part of a comprehensive information management system. The challenge is to build a single coherent information agent incorporating the functionality required to meet all of a user's information requirements.

Another area for ongoing research is representing and reasoning about multimedia information objects, such as images and video, and audio media. This has been touched

Table 2 Comparison of information management systems.

	Automatic query generation	Query enhancement	Use of user profile	Profile adaptation	Information extraction	Automatic information sharing	Automatic information categorisation	Proactive information management
Jasper			•	•	•	•		•
ProSearch		•	•	•				•
nDir			•				•	
RADAR	•			•				•
ProRate			•	•		•	•	•

upon in section 4.3 where ProRate was described, a system which can capture user preferences about non-text objects implicitly (that is, without the requirement to tag or index the objects in some way). The retrieval, management and organisation of non-textual information remains an area where much research remains to be done.

In this paper, an information management process has been set out which it is believed an information agent should ideally support. A number of novel systems developed at BTL which go some way to meet this process have been described. These systems represent steps along the road to the goal of producing an information agent which can meet all of a user's information needs in a timely and effective way.

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